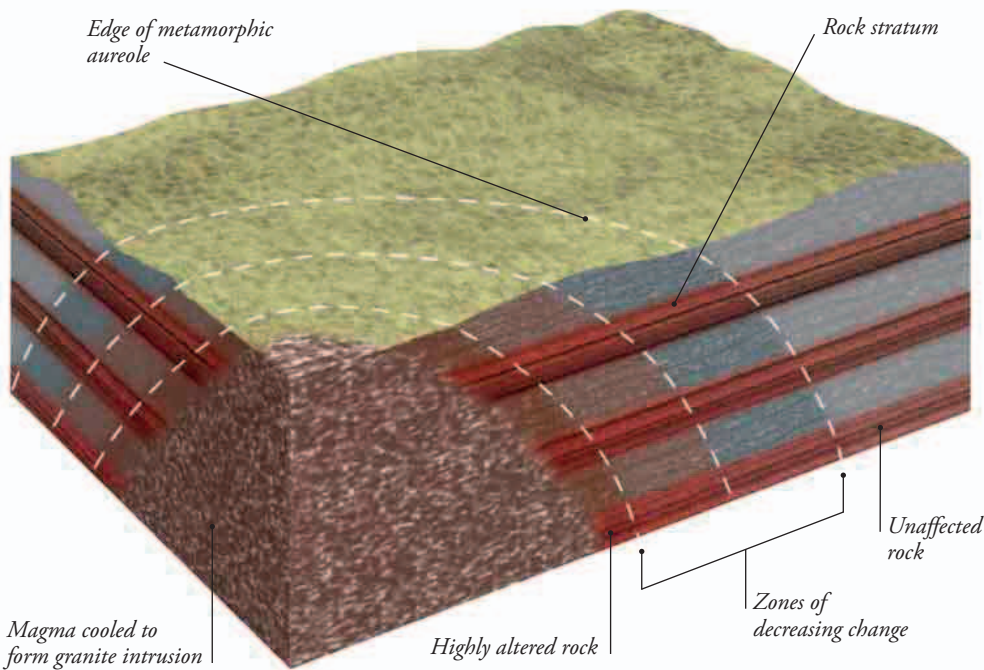


ROASTED ROCK

Upwelling magma that cools below ground cooks the rocks around it. The result is hornfels, a hard rock that may be spotted with mineral crystals. The areas where it is found, around igneous intrusions, are known as metamorphic aureoles. They might also contain veins of metal ore yielding copper, lead, and silver.



FAST FACTS

- Many metamorphic rocks are formed by high temperatures of 900°F (500°C) or more—almost high enough to melt them.
- Metamorphosed sandstone turns into an extremely hard rock called quartzite, which is virtually pure quartz.
- Although marble is durable, it can be dissolved by lemon juice and even by acid rain, which damages marble stonework.
- Precious metals and gemstones are often created by the processes that form metamorphic rocks.

◀ HEAT TREATMENT

A mass of hot magma pushed up into existing rocks cooks and alters them.

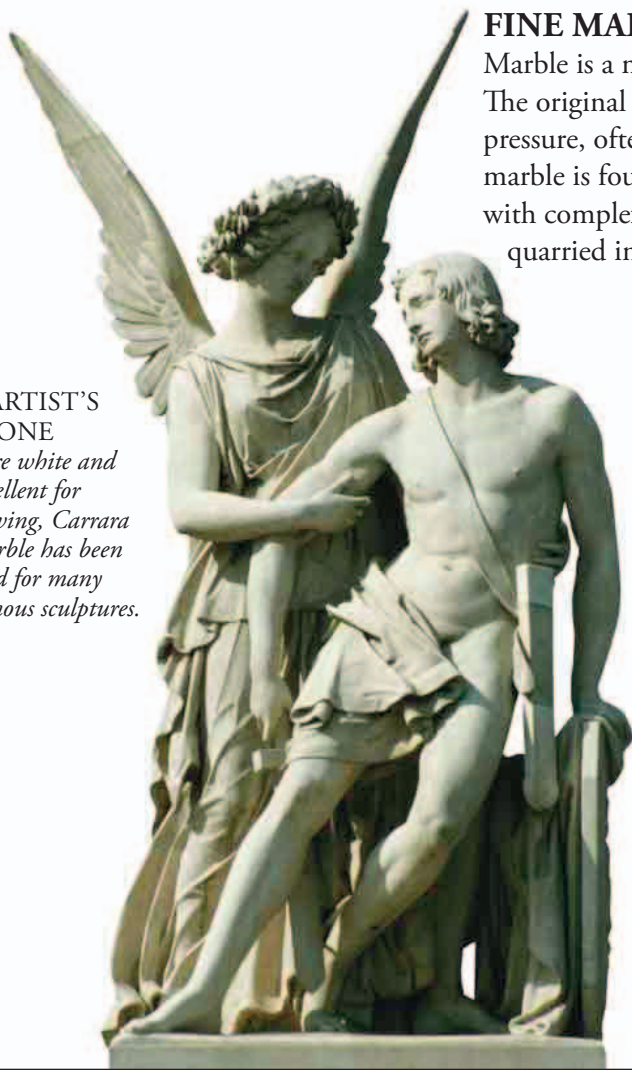
FINE MARBLE

Marble is a metamorphic rock formed from chalk or limestone.

The original rock is recrystallized by high temperatures and extreme pressure, often during periods of mountain building. This is why most marble is found in mountain ranges. Marble can be many colors, often with complex patterns, but some marbles, such as the Carrara marble quarried in Tuscany, Italy, are pure brilliant white.

► ARTIST'S STONE

Pure white and excellent for carving, Carrara marble has been used for many famous sculptures.



DEEP HEAT

Before they were changed by heat or pressure, or both, most metamorphic rocks were sedimentary rocks such as shale, sandstone, or limestone. But igneous rocks can undergo the same process. They include the mantle rock peridotite, which can be transformed into serpentinite and eclogite. These form when rock is put under pressure deep in the subduction zones between the colliding plates of Earth's crust.

► MANTLE ROCK

This serpentinite, found in Cyprus, formed from rock in the oceanic crust.

